Sri Venkateswara College of Engineering

Data Plane and Control Planes

Core Components of Network Devices

Quick Recap: SDN Overview

- What is SDN?
- Separation of Control Plane and Data Plane
- SDN architecture layers: Application, Control, and Data
- Centralized vs Distributed Control/Data

Introduction

- Network devices contain two crucial logical layers:
  - The Data Plane (Forwarding Plane)
  - The Control Plane
- Both play distinct roles but are highly interconnected.

Data Plane Overview

- Also known as the Forwarding Plane
- Primary responsibility: Efficient data forwarding
- Handles: Packets, frames, or datagrams
- Performs: Basic sanity checks

Key Functions of the Data Plane

- Packet Processing & Forwarding: Uses the Forwarding Information Base (FIB) for fast, rule-based forwarding
- Actions on Traffic:
  - Forwarding: Send to outgoing port(s)
  - Dropping: Discard packet
  - Re-marking: Modify header fields
  - Counting: Update traffic stats
  - Queuing: Buffer for later forwarding

Data Plane Implementation

- FIB Table Locations:
  - Software
  - Hardware-accelerated (e.g., CPU/GPU)
  - Commodity silicon (NPU)
  - FPGAs
  - Specialized silicon (ASICs)
- Local Services:
  - Access Control Lists (ACLs)
  - Quality of Service (QoS) policies

Data Plane – "Punt Path"

- When unmatched traffic or control packets (unknown destinations or special control packets) arrive:
  - Sent (“punted”) from Data Plane to Control Plane
  - Low-throughput, reserved for special cases (e.g., unknown destinations, routing protocol packets)

Data Plane in SDN Context

- SDN Evolution: Data plane devices become “dumb” (but fast) forwarding elements
- Total behavior dictated by flow rules set by an external, centralized SDN controller
- Communication via southbound APIs, e.g., OpenFlow

Control Plane Overview

- Acts as the “network brain”
- Builds a consistent view of network topology
- Responsible for programming the FIB used by the data plane

Key Functions of the Control Plane

- Network View Construction:
  - Manual configuration
  - Passive observation
  - Communication with other devices (routing protocols like OSPF, BGP)
- Maintains the Routing Information Base (RIB)
- Updates FIB entries in line with current topology
- Handles unmatched/control traffic: Learns info, updates RIB/FIB, and resends as needed

Control Plane in Traditional vs. SDN Networks

Aspect	Traditional Networks	SDN Approach
Coupling	Integrated with data plane	Usually decoupled
Location	Distributed in every device	Centralized in a controller/NOS
Flexibility	Manual, decentralized	Automated, programmable, global view

SDN Controller / NOS (Network Operating System)

- Locally decouples the control logic
- NOS provides:
  - Topology/State abstractions
  - Device management services
  - APIs for application/developer use
- Enables:
  - Easier policy programming
  - Automated network adaptation
  - More advanced network services

Summary

- Data Plane: High-speed forwarding, simple logic, actions dictated by rules/programming.
- Control Plane: Makes forwarding decisions, adapts to topology, manages rules.
- SDN: Separates these planes for superior programmability, scalability, and network innovation.